11/10 ~~Lecture~~ colloquium Jessie Muir

Big DESI Person

- Cosmology → using large-scale universe to learn fundamental physics
- We infer the existence of Dark matter and Dark energy by how they interact with visible matter
- Cold dark matter → non relativistic
- Dark energy could be time varying
- Distance measurements give expansion history
- The BAO scale sounds neat
- if not constant Λ allow ρ_{DE} to vary or EoS to vary with time.
- DESI suggests for both a time varying EoS and ρ_{DE}
- Imaging surveys probe physics
  - Background expansion & structure growth
  - Galaxy clustering & supernovae help study expansion
  - galaxy clustering, clusters & lensing help study structure

[Why does DES see so few galaxies/arcmin^2
6 vs 30 arcmin in LSST]

- The Ω_Λ Ω_m relation led to ΛCDM
- modeling is extremely important

(cosmic shear, galaxy clustering & lensing have different systematics so results should check each other

Ivo main talk continued

- hard to calibrate against multiple bias
- constrains ~~Phot z params to~~ cosmo params by comparing predictions to 2pt measurements
  - Bayesian analysis.
- A handfull of sources of error
- Tension between early universe and late universe on amplitude of density functions S_8 and Ω_M
  - ΛCDM does a pretty good job
- DESI results have been gaining precision and find an S_8 value $\sim 1\sigma$ lower than Plank
- LSST will allow for more exciting analysis
- lots of cool statistics using concepts I'm learning in 457
- cosmology is a modeling problem
  - fitting data to models

- What exactly is 2pt analysis?